
OBAFEMI AWOLOWO UNIVERSITY

ILE-IFE, NIGERIA



FACULTY OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSC 514 COMPUTER GRAPHICS

**Rain Semester Examination
2022/2023 Academic Session
July, 2024**

INSTRUCTIONS

TIME ALLOWED: 2 Hours

**ANSWER ALL QUESTIONS IN SECTION A AND ANY THREE (3)
QUESTIONS FROM SECTION B**

SECTION A (25 marks)

Fill in the gaps.

1. Smallest size object that can be displayed on a monitor is called _____.
2. On a monochromatic monitor, the frame buffer is known as _____.
3. The distance from one pixel to the next pixel is called _____.
4. An RGB color system with 24 bits of storage per pixel is known as _____.
5. Identify different type of computer graphics
6. _____ is responsible for accessing the frame buffer to refresh the screen.
7. Digitizing a picture definition into a set of intensity values is known as _____.
8. Graphics and image processing technique used to produce a transformation of one object into another is called _____.
9. Vector graphics is composed of _____.
10. Raster graphics are composed of _____.
11. EPS image file format is used for _____ and _____.
12. TIFF (tagged image file format) are used for _____.
13. Two dimensional color model are _____ and _____.
14. RGB model are used for _____.
15. CMYK model are used for _____.
16. The simplest output primitive is _____.
17. A bitmap is _____ bit(s) per pixels.
18. Each pixel has _____ basic color components
19. The quality of an image depend on _____.
20. The basic geometric structures that describe a scene on display is called _____.
21. The transformation in which an object is moved in a minimum distance path from one position to another is called _____.
22. The translation distances (dx, dy) is called as _____.
23. The transformation in which an object is moved from one position to another in circular path around a specified pivot point is called _____.
24. The transformation in which the dimension of an object are changed relative to a specified fixed point is called _____.
25. The point about which an object is rotated is called _____.

SECTION B**Question #1 (20 Marks)**

- a) What are the functions of the following in OpenGL?
 - i. `glBegin(GL_TRIANGLES);`
 - ii. `glColor3f(1.0,0.0,0.0);`

- iii. `glVertex3f(20.0,20.0,0.0);`
 - iv. `glTranslatef(5.0,0.0,0.0);`
 - v. `glRotatef(45.0,0.0,0.0,1.0);`
 - vi. `glutWireCube(5.0);`
 - vii. `glutCreateWindow()`
- b) Write the 3x3 matrix corresponding to each of the following transformations
- i. Translate by the displacement vector $[-2,3]$
 - ii. Scaling by a factor of 2 in the x direction and 4 in the y direction.
 - iii. Counter-clockwise rotation by an angle of -45° about the origin.
- c) Explain the differences between Random and Raster Scan Displays.

Question #2 (20 Marks)

- a) Discuss the applications of Computer Graphics
- b) What do you understand by OpenGL primitives and explain the functions of the following set of statements/commands in OpenGL
- i. `GL_LINE_STRIP`
 - ii. `GL_LINE_LOOP`
 - iii. `GL_TRIANGLE_FAN`
- c) Explain the following terms in computer graphics.
- i. aliasing ii. Rendering iii. Coordinate System iv. Z-buffer
 - v. Dithering vi. jaggies

Question #3 (20 Marks)

- a) Explain what homogeneous coordinates are and why we use them? How can they be used in computer graphics to model (i) translation and (ii) simple perspective
- b) What is a point, a circle and ellipse in the computer graphics system?
- c) Write short notes on active and passive transformation? And explain the following terms
- i. Translation ii. Rotation iii. Scaling iv. Shearing and v. Reflection

Question #4 (20 Marks)

- a) Write the equations that convert a color from the RGB model to CMY Model.
 - i. Find the RGB coordinates of a color at (0.75,0.95,0.4) in the CMY space.
 - ii. Write the pseudocode to implement this conversion.
- b) Explain the difference between additive and subtractive colour mixing.
- c) Give three common applications of vertex shaders in OpenGL.